

Object: Subject of your organization's *Policy-in-Action* projects 2025-2026

Dear Partner,

In preparation for the 2024-2025 academic year, please let us know by **September 15, 2025**, of one (or more) "*Policy-in-Action*" topic(s) you would like to propose to our students. The expected format is shown below.

As a reminder, a "Policy-in-Action" project consists in proposing to our students a subject and/or a problem that your teams are addressing. Each project will be handled by a group of 3 to 5 students and will be **supervised by us**. Projects will be monitored on an ongoing basis, and all we expect from you is to take part in milestones, typically: an initial kick-off and scoping meeting (1h), a follow-up meeting where students can show you interim results and ask you questions about problems encountered (1h), and a final meeting where results will be presented to you (1h). At the end of this work, the programs and analyses carried out during the project will be made available to you, and where appropriate valorized on a GitHub and deployed on Onyxia.

The provisional timetable is as follows:

October 2025:	Kick-off and scoping meeting
December 2025 / January 2026:	Intermediate meeting
March 2026 / April 2026:	Final feedback meeting

We look forward to receiving your proposals, which we hope will be numerous, and remain at your disposal should you have any questions or wish to receive feedback on your topics.

Yours sincerely

Program directors Franck Malherbet and Pierre Boyer



Policy-in-Action Project proposal: year 2025-2026 (Return
by 15/09/2025)

Topic / Title: Shortages of medical staff in emergency departments and unannounced closures

Presentation of the subject proposed to the students (one paragraph):

The problem of a shortage of emergency physicians will still be a major issue in 2025, even though the number of emergency physicians is higher than it was ten years ago, even when compared with the number of emergency room visits: in ten years, the number of emergency room visits has risen by 9.1% (from 19.6 million in 2014 to 21.3 million in 2024), while the number of physician positions has increased by almost 20% (from 5,300 annual full-time equivalents [FTEs] in 2014 to 6,500 reported FTEs in 2021, then from 6,100 average annual FTEs of salaried staff alone in 2022 to 6,000 salaried FTEs in 2024). Is it simply that these medical resources are unevenly distributed? Our task will be to identify and characterize the departments where the ratio of doctors to passages (or the number of doctors) is insufficient; and, more fundamentally, to describe and characterize the importance of the shortage of emergency physicians in the partial or complete closures of emergency departments (bearing in mind that part of the task will be to establish an exhaustive database of these closures, by identifying the appropriate sources or using LLM models).

On a purely complementary basis, the same type of analysis could also be carried out on maternity units, and the ratio of births per number of obstetricians or midwives, as well as temporary or permanent maternity unit closures.

Problematic and technical issues (a few paragraphs):

Students will first be able to use the emergency department data from the [Statistique Annuelle des Etablissements de santé](#) (SAE, a comprehensive annual survey of healthcare establishments in France), to identify and characterize departments where the ratio of doctors per patient (or the number of doctors) is insufficient. As the data on emergency room visits in these reports is highly controlled, but the data on emergency room staff is somewhat less so, this will already be an opportunity to critically analyze survey data (and even to propose statistical adjustments in the most obvious cases). It will also provide an opportunity for descriptive statistical work in cross-sections (2024; 2014) to understand where there are few medical staff (in absolute terms, or relative to the number of visits), and to analyze this according to different dimensions (legal sector, type of department or urban area, general/adult/pediatric emergencies, size of department in terms of number of visits, etc.), or in panels (in which places has the situation deteriorated in ten years, and how are they characterized?).

The Emergency survey could be used at this point, as it offers a detailed characterization of services in 2013 and 2023, with the situation on a given day of the year in 2023 (and according to three time slots for staff), which could offer a counterpoint to the situation over one year presented by the SAE.

A complementary part of the students' work will then consist in automatically establishing a database of partial closures of emergency services and their modalities (according to which periods, closing times) and the implementation of prior regulation (according to which periods, at which times) since at least 2023. To do this, they will be able to consult the website of the Regional Health Agencies (for example, [here](#) for Brittany in 2024/2025) or contact them, and above all use structured data collection via an LLM model (with Internet query) to establish this (see [here](#) an example - without structuring the response and without automation - of an LLM request with Internet query showing that this is possible in this context). The completeness and error rate of this data collection can be compared with several sources: the data collected by the ARS (Regional Health Authorities) on the one hand; the Urgences 2023 survey, structure section, for the period prior to June 2023 on the other hand; daily data from RPU by structures on the other hand, for which an agreement will allow confidential access to students. As a possible extension, this collection could also include the reason given for the closure, a reference, and, beyond closures, the reported risks of closure of emergency facilities.

Once this basis has been established, and made credible through cross-checking, the students will seek to establish whether the establishments with such closures or regulations overlap with the establishments with few medical staff identified in the first part; if this is not the case, it will be interesting to determine other risk factors (legal sector? distance from the nearest emergency department? etc.). A statistical description of closures and regulations, from a regional angle and in terms of trends, would also be very interesting (see in particular Khaoua, with Suarez Castillo [2024]).

Bibliography or indicative references :

You can indicate what may have already been done by your teams, recommended documentation and suggestions for academic and/or institutional reading. As students are not necessarily French-speaking, please choose sources in English.

- [La Statistique annuelle des établissements de santé \(SAE\) | Direction de la recherche, des études, de l'évaluation et des statistiques](#)
- [Urgences 2023 survey | Direction de la recherche, des études, de l'évaluation et des statistiques](#)
 - Demoly E., Deroyon T. (2025, March). [Urgences : la moitié des patients y restent plus de 3 heures en 2023, 45 minutes plus qu'en 2013](#). DREES, *Études et Résultats*, 1334.
 - Demoly E., Bara W., Naouri D., Boisguerin B., Bianchi C. (2024, July). [Urgences hospitalières en 2023: quelles organisations pour la prise en charge des patients?](#) DREES, *Études et Résultats*, 1305.
- Khaoua H., with the collaboration of Suarez Castillo M. (2024, December). [Passages aux urgences entre 2017 et 2023 : des dynamiques contrastées selon les départements](#). DREES, *Études et Résultats*, 1320.

Targeted solution:

See above.

Data made available or available in open data:¹

Policy-in-Action Projects are applied projects. As far as possible, it is important to consider data access in advance. DEPP students are by default enrolled in CASD. If your project requires CASD access, please let us know. It is also possible to make data available on [Onyxia Datalab](#) and to draw up a framework agreement.

SAE (open data)

Emergency survey (structure section in open data; HR section not yet in open data).

RPU daily data by structure (by adapting last year's agreement to this subject).

Software / Software solutions :

To facilitate the appropriation and use by your teams of the solutions proposed within the framework of the project, you can, if necessary, indicate your preferences for the software solutions used. Our students all have skills in R, Python and Stata.

Very strong preference for R. We understand if Python is used for interaction with LLMs.

Who will use the results of the project and the tool developed?

The results and statistics will be of great interest to the Bureau Etablissements de Santé (BES).

The collection of closures and regulations will be developed from a retrospective perspective (no need to look at 2025 at this stage); a further development could be to modify it to make it a real-time monitoring tool, in which case it could probably be of interest to colleagues at the Direction Générale de l'Offre de Soins (DGOS) in charge of these subjects.

Do you plan to offer an end-of-year internship following on from this project?

As a reminder, DEPP students must do a long internship of at least 16 weeks at the end of M1 and at least 20 weeks at the end of M2. Internships generally run from the beginning of April to the end of August.

Depending on the results of the project (and the internship credits available), an internship could be envisaged to deepen one of the subjects and carry out a Studies & Results.

Do you have any additional comments?

Names and contact details of the people in your organization involved in this project:

Referent 1 :

Last name: Cazenave-Lacrouz

First name:

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¹For Policy-in-Action projects, confidential and non-confidential data may be processed on ENSAE's Onyxia platform (<https://www.onyxia.sh/>). Only students duly authorized to work on the project will be able to access the data.

Referent 2 :

Last name : First
name :

Email: Phone:

Commented [AC1]: This may be enriched if the subject is chosen by students.

